

# Modulation of miR23b Wnt/catenin Axis Strengthens Endothelial Barrier Properties

1 Introduction Neurological disorders with a dominant vascular component, such as stroke and Alzheimer's disease (AD), impose a profound healthcare burden. These conditions feature heterogeneous pathobiology, driven by diverse risk factors that frequently stem from bloodbrain barrier (BBB) impairment [ 1 , 2 ]. With over 15 million cases annually, stroke is a leading cause of death and longterm disability, and most cases are ischemic [ 3 , 4 ]. Early BBB disruption is a critical factor driving poor prognosis [ 5 , 7 ]. As a major risk factor, aging further compounds this issue through preexisting BBB dysfunction [ 8 , 9 , 10 , 11 , 12 , 13 , 14 , 15 , 16 , 17 ]. Changes in blood plasma protein composition resulting from transcriptional changes in brain endothelial cells (BECs) and systemic chronic inflammation are thought to contribute to agerelated BBB dysfunction [ 18 , 19 , 20 , 21 , 22 , 23 ]. Additionally, breakdown in BBB integrity is now recognized as a precursor to cognitive deficits, serving as an early, measurable marker for AD and dementia risk [ 12 , 13 , 14 , 15 ]. The BBB is a highly selective semipermeable barrier that regulates the passage of substances between the blood and the brain. Its complex, dynamic structure comprises tight junctions (TJs) composed of proteins such as Claudin5, Occludin, and ZO1 in BECs, along with a low rate of caveolar transport. Supporting cells, including pericytes, astrocytes, and the capillary vascular basement membrane, contribute to the structural and functional architecture of the BBB [ 24 ]. Despite growing insights into the BBB in CNS pathology, significant knowledge gaps persist regarding the spatiotemporal mechanisms underlying its dysfunction in ischemic stroke and the development of effective targeted therapies. The Wnt/catenin pathway and downstream transcription factors are critical for CNS angiogenesis, proliferation, survival, and BBB formation through stabilization and nuclear translocation of catenin [ 24 ]. Dysregulation of this pathway has also been implicated in stroke, AD, and MS [ 25 , 26 , 27 , 28 , 29 ]. More recently, small noncoding RNAs that modulate posttranscriptional gene expression, microRNAs (miRs), have emerged as important regulators of BEC function and BBB integrity [ 30 , 31 , 32 ]. miRs exemplify the growing recognition that noncoding RNAs possess a regulatory significance comparable to that of proteins. miRs are incorporated into Argonaute proteins to function as guide RNAs, directing the complex to target mRNAs through partial base pairing. Target recognition is partly governed by a short sixnucleotide seed sequence (positions 27), which binds complementary sites in target mRNA and results in posttranscriptional silencing by either reducing mRNA stability or suppressing protein translation [ 33 , 34 , 35 , 36 ]. We and others have shown that miRs can be induced by cytokines such as interferon (IFN) and tumor necrosis factor (TNF), and that they play critical roles in modulating inflammation [ 37 , 38 , 39 , 40 ]. With approximately 70% of miRs expressed in the mammalian brain, these molecules are critical for maintaining CNS homeostasis and are implicated in various neurological disorders. Consequently, miRs are emerging as promising nextgeneration therapeutic targets for CNS diseases. Here, we employ a nonbiased antimiR lentiviral screening strategy and identify antimiR23b/miR23b as key regulators of BBB function. Leveraging gain and lossoffunction approaches and multiomics, this study examines the mechanisms by which antimiR23b/miR23b regulates BBB function in healthy versus ischemic stroke conditions, using 2D and 3D microfluidic in vitro platforms, validated in an in vivo mouse model for stroke (tMCAO). Our data demonstrate that antimiR23b enhances BBB properties by regulating junctional proteins, transcytosis, and key signaling pathways in the brain endothelium, most notably Wnt/catenin, TGF, VEGF, and Notch. By improving BBB integrity, this approach significantly ameliorates outcomes in ischemic stroke models. These findings have high physiological and clinical relevance, positioning miR inhibitors as promising therapeutic agents.

2 Results

2.1 Identification of antimiR23b as an Enhancer of Endothelial BBB Properties miRs are key regulators of brain homeostasis and are frequently dysregulated in neurological disorders, making them attractive therapeutic targets [ 41 , 42 , 43 , 44 , 45 ]. To identify critical negative regulators of cellcell contact and barrier integrity, we performed an unbiased lentiviral screen using an antimiR library targeting an array of conserved miRs, by measuring transendothelial electrical resistance (TEER) as a functional readout. This approach uses antimiRs to neutralize endogenously expressed miRs, minimizing artifacts associated with ectopic overexpression. We previously validated this strategy by identifying miR128 as a key regulator of reverse transcriptaseassociated pathways, including LINE1, telomerase, and HIV1 [ 46 , 47 , 48 , 49 , 50 ]. Primary screening and validation were conducted using the following methodology. Primary mouse lung endothelial cells (ECs) were transfected with the antimiR library to enable highthroughput lossoffunction evaluation of miR candidates. Viral expression was selected using G418 (Neomycin) resistance and validated by copGFP expression (characterized by superbright green fluorescence). Infected ECs were clonally expanded in 96well plates. Once EC clones reached confluence, barrier function was assessed using an electric cellsubstrate impedance sensing (ECIS) instrument to measure TEER. EC wells showing a significant increase in TEER were harvested. The specific integrated antimiR sequences were identified by cloning and validation (Figure 1A and Figure S1A ). Three clones showed significantly increased TEER compared to controls (Figure 1B ), and three antimiRs were independently isolated from at least two different clones. Among these, antimiR151a and antimiR375 were identified as enhancers of cell junctions. We and

others have previously shown that miR151a and miR375 play roles in tumorigenesis, and that miR151a functions as an oncogenic miR to promote cell migration, invasion, partial epithelial to mesenchymal transition in non-small cell lung cancer, and tumor-associated angiogenesis [ 51 , 52 ].

**FIGURE 1 Identification of anti-miR23b as a promoter of barrier properties in brain endothelial cells.** (A) Mouse lung endothelial cells were transduced using a pooled lentiviral-based, miR-neutralizing shRNA library, selected for puromycin resistance, and clonally expanded. Each well represents the neutralization of a single endogenously expressed miR. Cells were then analyzed by transendothelial electrical resistance (TEER) by employing an Electric Cell Substrate Impedance Sensing (ECIS) plate reader to examine paracellular barrier properties. (B) Single high-titer miR-control, miR23b, and anti-miR23b lentiviruses were generated and stable miR-modulated human lung endothelial cells established. TEER was measured to assess the specific and opposing effect of miR23b and anti-miR23b,  $n = 12$ . (C) miR expression profiling was performed by miR-specific qPCR analysis of mouse brain endothelial cells (BECs) from embryonic and adult brains as well as primary human brain microvascular endothelial cells (HBMEC) and shown as a heatmap,  $n = 6$ . (D) Human HUVECs were transfected with miR control or anti-miR23b mimics oligonucleotides, and (E) stable HBMEC (parental, miR control, and anti-miR23b) lines were generated. HUVEC or HBMEC cells were subjected to TEER by ECIS or EVOM3 analysis,  $n = 4$ . Results are shown as percent change Mean  $\pm$  SEM of  $n = 412$  independent biological replicates, \*  $p < 0.05$ , \*\*  $p < 0.01$ , and \*\*\*  $p < 0.001$  by ordinary one-way ANOVA at each time point. Importantly, this screen also identified anti-miR23b as a potent enhancer of endothelial barrier properties. Subsequently, high-titer lentiviruses encoding miR23b and anti-miR23b were generated. Human lung ECs were infected with lentiviruses expressing control miR, miR23b, or anti-miR23b and subjected to secondary TEER analysis. This confirmed that miR23b and anti-miR23b exert opposing effects on EC TEER measurements, establishing their role in regulating cell-cell contact and barrier integrity in lung ECs (Figure 1C ). To assess the relevance of the miRs identified in our screen to brain EC barrier properties, we measured the expression of several candidates in mouse and human brain endothelial cells (BECs) using miR-specific qRT-PCR. miR375 was barely detectable in young and old mouse BECs and in primary human brain microvascular endothelial cells (HBMECs), whereas miR151a showed low expression in young mice but robust expression in adult mice and HBMECs. In contrast, miR23b was highly enriched across all cell types, including young and aged mouse BECs as well as HBMECs (Figure 1C ). This aligns with previous studies that have highlighted miR23b enrichment in the vasculature [ 53 ]. miR23b is part of the miR232724 cluster, with two paralogous clusters (miR23a and miR23b) arising from ancient gene duplication events on human chromosomes 9q22 and 19p13. miR23a and miR23b share an identical seed sequence but differ by one nucleotide [ 54 ], and both are expressed in the brain endothelium alongside miR27 and miR24 (Figure 1C ) [ 53 ]. Given these expression patterns, miR23b/anti-miR23b were selected as the primary focus of this study. To determine the functional role of anti-miR23b in endothelial barrier integrity, we evaluated its effects on human umbilical vein endothelial cells (HUVECs) and HBMECs. In both cell models, anti-miR23b administration significantly increased TEER compared with control cells (Figures 1D,E ). These findings indicate that miR23b plays a critical role in modulating paracellular barrier permeability across endothelial lineages.

## 2.2 Silencing miR23b Enhances the Barrier Properties of HBMECs by Upregulating Cell Junction Proteins

Selective BBB permeability depends on specialized adherens (AJ) and tight junctions (TJ) and their associated proteins, including VECadherin, Claudin5, Occludin, and ZO1 [ 55 , 56 ] (Figure 2A ). We investigated whether miR23b and anti-miR23b regulate the expression of these junctional proteins in BECs. **FIGURE 2 anti-miR23b promotes barrier properties through increased expression of cell junctional proteins.** (A) Cartoon of key junctional proteins in brain endothelial cells. miR-modulated HBMEC or hCMEC/D3 cells were generated (overexpressing anti-miR23b, miR23b, or control miR), or HUVECs were transiently transfected with anti-miR23b, miR23b, or control miR oligonucleotides. Stable and transiently transduced anti-miR23b, miR23b, and control miR cell lines were analyzed for changes in protein levels of key tight junction proteins (TJPs; ZO1, Claudin5, and Occludin) by Western blot (WB) analysis. (B) Representative WB examples of anti-miR23b relative to control miR hCMEC/D3, HBMEC, and HUVEC are shown (left panel), and quantification of WB results normalized to GAPDH are shown as relative levels,  $n = 3$  (right panel). (C) Representative WB examples of miR23b and controls hCMEC/D3 and HBMEC are shown (left panel), and quantifications are shown as relative levels,  $n = 3$  (right panel) (right). (D) Representative immunofluorescence (IF) images of transduced primary HBMECs and transiently transfected HUVECs stained for ZO1 (green) and Claudin5 (red) or Occludin (red, left panel), and quantification of relative fluorescence levels,  $n = 3$  (right panel), scale = 50  $\mu\text{m}$ . (E) Stable anti-miR23b and control miR cell lines were analyzed for expression levels of the adherens junction protein VECadherin by IF (left panel), WB (middle panel), and quantifications,  $n = 3$  (right, and far right panels), Scale = 25  $\mu\text{m}$ . (F) Results of 3'UTR luciferase reporter assays for junctional proteins is shown. HEK293 cells were cotransfected with 3'UTR containing plasmids for TJP1, CLDN5, OCLN, CDH5 or control, with miR control or miR23b mimic oligonucleotides and luciferase activity was measured after 24 hrs,  $n = 4$ . Graphs are shown as Mean  $\pm$  SEM (  $n = 34$ , independent biological replicates), \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , by two-way ANOVA for multiple

comparisons and unpaired student's t test when comparing two groups. We established miR23b overexpressing, anti-miR23b, or control miR cell lines in both HBMECs and the human cerebral microvascular endothelial cell line (hCMEC/D3) by lentiviral transduction (routinely achieving 90% transduction efficiency). Concurrently, HUVECs were transiently transfected with miR23b, anti-miR23b, or control miR mimic oligonucleotides. Protein levels of ZO1, Claudin5, and Occludin were then assessed by Western blot (WB) analysis. Anti-miR23b treatment markedly increased the levels of all three TJ proteins across all three endothelial cell models (Figure 2B), whereas miR23b overexpression diminished their expression compared to control cells (Figure 2C). Building on these findings, we conducted immunofluorescence (IF) staining to determine the subcellular localization of these proteins in HUVECs and HBMECs. In both cell types, anti-miR23b overexpression promoted robust localization of Claudin5, Occludin, and ZO1 at cell-cell junctions (Figure 2D and Figure S2A), indicating improved barrier integrity. Furthermore, IF and WB analyses revealed that anti-miR23b significantly upregulated junctional VECadherin levels in HBMECs compared to controls (Figure 2D). To determine whether miR23b directly targets the mRNAs of endothelial junctional proteins or regulates them indirectly, we conducted bioinformatic seed sequence analyses using TargetScanHuman 8.0 and 3'UTR luciferase reporter assays for ZO1, Occludin, Claudin5, and VECadherin. Bioinformatic analysis revealed a conserved 8mer seed binding site for miR23b within the 3'UTR of TJP1/ZO1 (Figure S2B). In contrast, no miR23b seed binding sites were identified in the 3'UTRs of CDH5, CLDN5, and OCLN. We next conducted luciferase reporter assays to determine whether miR23b directly binds these 3'UTR regions to suppress luciferase activity. 3'UTR luciferase constructs of the four junctional genes were cotransfected with miR23b or control miR mimics oligonucleotides into HEK293T cells. As shown in Figure 2F, miR23b significantly, albeit modestly, bound to TJP1/ZO1 and CDH5/VECadherin mRNAs, whereas no binding was detected with the 3'UTRs of CLDN5/Claudin5 or OCLN/Occludin. These findings suggest that anti-miR23b strengthens the endothelial paracellular barrier by upregulating tight and adherens junction proteins through both direct and indirect mechanisms. Although earlier studies established that various miRs regulate junctional proteins in cell types such as lung epithelium and endothelial cells [52, 57, 58, 59], our data demonstrate that inhibiting miR23b in BECs is critical for promoting barrier maturation under physiological conditions.

### 2.3 Anti-miR23b Enhances 3D Tubule Formation and BBB Properties of HBMECs

While traditional 2D endothelial monolayers offer limited physiological relevance for studying the BBB, 3D systems incorporating flow more accurately replicate *in vivo* vascular architecture and function. These advanced models significantly affect the expression of proteins critical to BBB integrity [60]. To better assess the impact of anti-miR23b on BBB integrity under dynamic 3D conditions, we employed a 3D HBMEC culture model using the Mimetas 40/64chip 3lane OrganoPlate microfluidic system (Figure 3A). This platform integrates tissue-level complexity with precise environmental control, enabling reproducible, quantitative assessments of BBB properties across multiple chips. Each chip uses only 30,000 HBMECs, enabling cost-effective feasibility and scalability. The OrganoPlate system facilitates gravity-driven fluid flow via platform tilting, creating physiologic shear stress critical for HBMEC maturation and function. Cells are seeded into the top channel (coated with Collagen IV (Col IV)/Fibronectin (FN)), and brain capillary structures form against an extracellular matrix (ECM) gel (Col I/IV, 80:20) in the middle lane (see Figure 3A, left, top panel). Functional, leak-tight 3D BBB tubules are visualized by performing barrier integrity (BI) assays using fluorescent dextran tracers of various molecular weights and colors, which are perfused into the upper channel to confirm the selectivity and maturity of HBMEC tubules. BI assays using a green 150 kDa dextran tracer (Figure 3A, right panel) and 3D expression of TJ proteins (Claudin5 and ZO1) demonstrate functional brain capillary-sized tubules (Figure 3B). **FIGURE 3 anti-miR23b accelerates 3D brain endothelial tubule formation and strengthens their barrier properties.** (A) Outline of the Mimetas organ-on-a-chip plate. Perfused 3D tubules are established in 40 or 64 3lane chip cultures using the Mimetas platform. The top channel is seeded with HBMEC against the ECM (middle channel), allowing for 3D tubule formation of brain capillary size, and the bottom channel can be seeded with CNS cells. A tight tubule only shows tracer in the top channel of the 3lane culture chip, whereas a leaky tubule will show tracer in the middle channel. Barrier integrity (BI) analysis is performed using different MW-sized tracers. Shown are intact HBMEC tubules in a 64chip plate using a 150 kDa FITC-dextran (green) tracer, and quantification of BBB permeability of a no cell chip culture showing complete leakiness (top) and an HBMEC chip showing no permeability, representing a functional BBB (bottom). (B) Intact 3D HBMEC tubule formation is visualized by ZO1 and Claudin5 staining. (C) The timing of tubule formation was examined using anti-miR23b and miR control HBMECs by BI analysis at 6, 18, and 30 h, n = 3. Scale = 250  $\mu$ m. (D) 3D TEER measures of parental, control miR, and anti-miR23b HBMECs were recorded, n = 12. (E) The proliferation potential of anti-miR23b and control HBMECs was determined by cell counting of triplicate cultures, using a cell coulter counter, n = 5. (F) Schematic of the 3D OrganoTEER equipment is shown (top, approved cartoon modification from Mimetas website), and plots of kinetic analysis of HBMEC tubule establishment (left panel) and long-term tubule resilience (right panel) were built from the TEER measurements, n = 1216. (G) Shown are IF staining for key junctional proteins [ZO1 (green), Claudin5 (red) or VECadherin (red), and nuclear stain (DAPI, blue)] in

antimiR23b and control 3D HBMEC tubules. Representative images of the full top channel tubule and magnified images (left panels) and quantifications are shown (right panel),  $n = 3$ , Scale = 250 and 25  $\mu\text{m}$ . All graphs are shown as Mean  $\pm$  SEM,  $n = 316$  independent biological replicates, \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$ , by twoway ANOVA for multiple comparisons and multiple unpaired ttests at each time point for TEER measurements. To determine how antimiR23b affects 3D vascular formation and barrier function in HBMECs, we compared antimiR23bexpressing HBMECs to control miR cells in the 3D assay. Both groups formed morphologically intact tubules; however, antimiR23b significantly accelerated the formation of leaktight HBMEC tubules, and tighter paracellular barrier function was observed as early as 18 h postseeding. In contrast, control tubules remained permeable at 18 h (Figure 3C ). This observation was also supported by 3D TEER measurements with the OrganoTEER, which showed significantly higher resistance in antimiR23b cultures than in parental and controlmiR HBMECs (Figure 3D ). Next, we tested whether antimiR23b affects HBMEC cell proliferation, observing a modest but significant increase in cell proliferation potential in antimiR23b HBMECs, compared to control HBMECs, as previously reported [ 63 , 64 ], likely contributing to the antimiR23b HBMECs enhanced barrier properties (Figure 3E ). Additionally, we tracked TEER during the first 24 h of tubule development to assess tubule establishment (Figure 3F , top). antimiR23b significantly accelerated the formation of functional tubules compared with controls (Figure 3E , left panel). Moreover, antimiR23b strengthened tubule resistance over three weeks in culture (Figure 3F , right panel). Finally, we examined expression levels of key junctional proteins by 3D IF staining for ZO1, Claudin5, and VECadherin using a C10 confocal imaging plate reader, demonstrating intact, lumenized 3D tubules in antimiR23b cultures with significantly elevated expression of ZO1, Claudin5, and VECadherin compared to controls (Figure 3G ). Confirming our 2D data (Figure 2 ), these findings demonstrate that antimiR23b enhances HBMEC barrier function and maturation in a physiological 3D context. Inhibiting miR23b accelerated and strengthened the formation of a 3D microfluidic BBB model, with robust junctional protein expression and sustained structural integrity.

#### 2.4 Inhibiting miR23b Enhances Transcellular Barrier Properties by Downregulating PLVAP

In addition to TJ integrity, low rates of transcellular transport are essential for maintaining BBB function [ 17 , 55 ]. Dysregulation of transcellular transport has been recognized as an early contributor to barrier breakdown following pathological insults such as stroke [ 61 , 62 ]. To assess whether antimiR23b modulates transcytosis in BECs, we examined levels of several proteins that regulate transcellular transport in antimiR23bexpressing and control HBMECs and hCMEC/D3 cell lines (see Figure 4A ). WB analysis revealed that antimiR23b significantly downregulated Plasmalemma VesicleAssociated Protein (PLVAP), a marker of fenestrated ECs (Figure 4B ). In contrast, Caveolin1 (Cav1) and Caveolin2 (Cav2, not shown) protein levels were not altered by antimiR23b expression (Figure 4B ). We performed a transcellular uptake assay using fluorescently labeled albumin (albuminAlexa594, red) to determine the functional consequences of PLVAP downregulation. After one hour of exposure, antimiR23b HBMECs showed a marked reduction in intracellular albuminAlexa594 signal compared to control cells, as shown in a panel of 3D HBMEC chip cultures (Figure 4C ), and in a representative 3D full tubule image at higher magnification, indicating reduced uptake (Figure 4D , left panel). Quantification confirmed a significant decrease in albumin uptake in antimiR23b cultures (Figure 4D , right panel). To independently validate this observation, we performed IF imaging of PLVAP and DAPI in antimiR23b versus control HBMEC cultures. antimiR23b expression led to a modest but statistically significant reduction in PLVAP protein levels (Figure 4E ), consistent with the WB findings. Therefore, in BECs, antimiR23b acts as a dual stabilizer of the BBB by reducing transcellular transport, via PLVAP downregulation, and enhancing the structural integrity of junctional proteins.

#### FIGURE 4 antimiR23b strengthens barrier properties by reducing transcytosis and downregulating PLVAP protein levels.

(A) Cartoon of brain endothelial cell junctional proteins and transcellular transport, which is regulated by PVLAP. (B) miRcontrol or antimiR23b overexpressing HBMEC or hCMEC/D3 cells were analyzed for changes in expression levels of Caveolin1 (Cav1) and PLVAP by Western blotting. Representative examples are shown (left panel), and quantification of results normalized to GAPDH as relative levels,  $n = 3$  (right panel). (C, D) Transcellular uptake of labelled Albumin594 (red, readout of transcellular transport), were assessed in control or antimiR23b HBMECs. (C) Albumin594 transcytosis is shown in multiple chip cultures,  $n = 5$ . Scale = 500  $\mu\text{m}$ . (D) A representative images of a single 3D tubule (left panel) and quantification of albumin transcytosis (right panel),  $n = 5$ . Scale = 25  $\mu\text{m}$ . (E) Confocal images of PLVAP/DAPI staining of antimiR23b or controls (left panel) and quantifications,  $n = 3$  (right panel), Scale = 12.5  $\mu\text{m}$ . Graphs are shown as Mean  $\pm$  SEM,  $n = 35$  independent biological replicates, \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$ , by twoway ANOVA for multiple comparisons and by unpaired twotailed Student's t test.

#### 2.5 Inhibiting miR23b Enhances Barrier Properties Through Wnt/CateninDependent and Independent Mechanisms

Wnt/catenin signaling plays a central role in promoting and maintaining BBB integrity [ 55 , 65 , 66 , 67 ]. CNS injury triggers reactivation of Wnt/catenin signaling, which contributes to vascular repair mechanisms [ 55 ]. Given this pathway's essential role in preserving the BBB, pharmacological or genetic modulation of Wnt/catenin represents a promising therapeutic approach in preclinical models

of neurological disorders, such as ischemic stroke [ 25 , 26 , 27 , 28 , 29 ]. To determine whether miR23b regulates the Wnt/catenin pathway, we first analysed protein levels of catenin and its transcriptional coactivator LEF1 in anti-miR23b-expressing versus control ECs. WB analysis revealed that anti-miR23b significantly increased the protein levels of catenin and LEF1 in hCMEC/D3, HUVEC, and HBMEC lines (Figure 5A ). We next corroborated these findings using IF in 2D cultures and 3D HBMEC tubules. There was increased expression of catenin and LEF1 in anti-miR23b HBMECs compared to controls in both 2D and 3D models (Figure 5B and Figure S4A,B ).

**FIGURE 5** anti-miR23b enhances barrier properties through Wnt/catenin dependent and independent mechanisms. (A) Levels for catenin and LEF1 proteins were examined by Western blot analysis in anti-miR23b or control HBMECs, hCMEC/D3 transduced cells, and in control or anti-miR23b mimic oligonucleotide transfected HUVECs. Results are shown as representative images (left panels) and quantified,  $n = 3$  (right panels). (B) Representative IF images of stable transduced HBMEC (control and anti-miR23b) stained for catenin (red) or LEF1 (green) in 3D tubules (left panel), magnification of catenin and LEF1 expression in HBMECs in 2D (middle panel), and quantifications (right panels),  $n = 3$ . Tubule scale = 25  $\mu\text{m}$ . Magnified image of catenin and LEF1 scale = 50 and 30  $\mu\text{m}$ . (C) WntGFP reporter assays using HEK293 cells transfected with M38TOPdGFP reporter plasmids with miR control or miR23b mimic oligonucleotides, using Wnt3a as a positive control,  $n = 6$ . (D) 3'UTR luciferase assays were performed for catenin (CTNNB1) and LEF1 by transfecting HEK293 cells with 3'UTR luciferase constructs for either CTNNB1, LEF1 or control with either control or miR23b mimic oligonucleotides,  $n = 7$ . Shown are results of junctional proteins and catenin and LEF1. (E) miR enhanced crosslinking and immunoprecipitation (miReCLIP) sequencing analysis, was performed for miR23b, see cartoon (top panel). Control, miR23b and anti-miR23b overexpressing HBMEC lines were generated, their barrier phenotype was validated by TEER analysis, and miR eCLIPseq was performed. The Table shows details of miR23b direct binding to VECadherin (CDH5), catenin (CTNNB1), and LEF1 mRNA. Underlined nucleotides in predicted seed, highlight mismatches. (F) Cartoon representation of likely miR23b effects in BECs (direct or indirect) on VECadherin, catenin, LEF1, ZO1, and Claudin5, Occludin, and PLVAP. (G) Global, unbiased overrepresentation analysis (ORA) was performed on HBMEC eCLIP results, showing enriched pathways as listed (left panel), and examples of miR23b mRNA targets for the top signaling pathways (Table, right panel). (H) To assess the contribution of Wnt/catenin signaling of the miR/anti-miR23b effect in BECs, rescue experiments were performed. To examine the effect of Wnt activation, control, miR23b or anti-miR23b HBMEC with Wnt3a (Wnt agonist) were challenged to TEER analysis, rescue effect shown in green (left panel). To examine the effect of Wnt inhibition, control, anti-miR23b or anti-miR23b HBMEC with XAV939 (Wnt inhibitor) were challenged to TEER analysis, rescue effect showed in red (right panel),  $n = 5$ . Graphs are shown as Mean  $\pm$  SEM,  $n = 37$  independent biological replicates, \*  $p < 0.05$ , \*\*  $p < 0.01$ , and \*\*\*\*  $p < 0.0001$ , by twoway ANOVA for multiple comparisons, oneway ANOVA when analyzing one variable, unpaired twotailed Student's  $t$  test for two groups, and multiple unpaired  $t$  tests when analyzing TEER measurements. Next, we performed WntGFP reporter assays to assess whether miR23b affects Wnt/catenin activity. Briefly, HEK293 cells were transfected with the M38TOP reporter along with control or miR23b mimic oligonucleotides, and Wnt3a served as a positive control for pathway activation. The assay revealed a significant reduction in WntGFP reporter activity in the presence of miR23b, demonstrating that miR23b inhibits Wnt/catenin signaling (Figure 5C ). To determine whether miR23b directly targets catenin and LEF1 mRNA, 3'UTR luciferase assays were performed as previously described. The results showed a modest but significant decrease in catenin 3'UTR reporter activity, with no direct binding to LEF1 mRNA. Reporter assay results for the junctional proteins and for catenin/LEF1 interactions are shown in Figure 2F and Figure 5D , respectively. Because miRs often bind with imperfect seed complementarity and can target mRNAs outside the 3' UTR, we used miR eCLIPsequencing to examine in-cell miR-mRNA interactions. By capturing chimeric RNA reads formed within the AGO2 complex, this approach directly maps miR23b-mRNA binding sites surpassing the limitations of computational predictions and reporter assays (Figure E, top panel). To define the direct targets of miR23b, control, miR23b overexpressing, and anti-miR23b overexpressing HBMECs were subjected to miReCLIPseq. Focused analysis of junction-related and Wnt/catenin pathway genes, including CDH5/VECadherin, CLDN5/Claudin5, OCLN/Occludin, TJP1/ZO1, CTNNB1/catenin, and LEF1, revealed that miR23b directly binds to CDH5, CTNNB1, and LEF1 mRNAs, showing more than threefold enrichment compared to controls (Figure 5E , Table, Figure S4C and Extended Data Set S1 ). Interestingly, these binding regions included nonperfect seed sequences, specifically within the 3' UTRs of CDH5 and CTNNB1 and the distal intron of LEF1 (Figure 5E ). These miR23b targets were successfully validated in hCMEC/D3 cells, lymphatic endothelial cells (LECs), or both (data not shown). These combined results establish that miR23b directly targets the mRNAs of CDH5, CTNNB1, and LEF1. miR23b may also bind TJP1 mRNA in certain cellular contexts, based on 3'UTR luciferase and bioinformatic analyses. However, Claudin5 and Occludin appear to be indirect targets of miR23b, likely regulated via the Wnt/catenin signaling pathway. As illustrated in the cartoon, anti-miR23b modulates this pathway to increase Claudin5 and Occludin expression while reducing PLVAP (Figure 5F ). Next, we performed an unbiased analysis of the miR23b eCLIPseq

datasets to examine global effects of miR23b/antimiR23b in HBMEC, identifying prominently altered pathways and further assessing the role of Wnt/catenin signaling. These analyses established that, in addition to CTNNB1 and LEF1, miR23b targets several other key mRNAs in the canonical Wnt/catenin pathway, including WNT7A, FZD4, LRP5, LRP6, APC, TCF3, TCF4, and DKK1. As shown in the OverRepresentation Analysis (ORA) dot plot, miR23b targets are enriched for numerous key signaling pathways, including the TGF pathway (TGF1, TGF2, TGFR1, TGFR2), VEGF (VEGFA, VEGFC, FLT1), Notch (NOTCH1, HEY1, DLL3), IL6 (IL6, IL6R, IL6ST), and the noncanonical Wnt/catenin pathway (WNT5A, ROCK1, DVL2, DVL3, DAAM1) (Figure 5G, left and Figure 5G Table, right). The complete miR eCLIP target list is available as Extended Data Set S1. Collectively, these findings indicate that antimiR23b elicits robust, multifaceted effects on BBB integrity. Specifically, the canonical Wnt/catenin pathway promotes structural maturation by enhancing tight junction formation and suppressing vesicular transport. Concurrently, noncanonical Wnt signaling modulates cell polarity and stress responses, and both pathways act in concert to fortify barrier stability. antimiR23b enhances BBB properties by promoting maturation of the neurovascular unit, specifically by increasing TJ protein expression via TGF and the MEK/ERK pathway. Notch signaling is a critical, conserved regulator of BBB integrity and cerebrovascular homeostasis, whereas VEGF and IL6 signaling typically increase BBB permeability, particularly following injury. However, under precise, lowdose conditions, VEGF and IL6 can promote protective angiogenesis and neurogenesis. The ORA analysis further revealed that miR23b targets numerous mRNAs and proteins critical for cellcell adhesion (CDH5, CTNNA1, PECAM1, JAM3), angiogenesis (NOTCH1, VEGFA, SNAI1, SERPINE1), BBB function (CLDN12, JAM3, CDH5, FOXP1, FOXO1), and the extracellular matrix (ECM) (COL4A1, COL4A2, LAMB1, LAMA2, LAMC1, ITGA2, ITGB3). Together, these effectors are essential for maintaining BEC function and BBB integrity in health and disease. Finally, to examine Wnt/catenindependent and independent mechanisms of miR/anti23brelated effects on BBB properties, we performed dual rescue experiments. First, we assessed Wnt/catenin activation by culturing miR control or miR23b HBMECs in the presence or absence of Wnt3a, a Wnt agonist, and measured the effect on 3D TEER. If the Wnt/catenin pathway is required for the antimiR23b protective phenotype, the Wnt agonist would be expected to inhibit the miR23b-induced loss of barrier integrity (Figure 5H, left panel in green). Second, we evaluated Wnt/catenin inhibition by applying a specific Wnt/catenin inhibitor (XAV939) to control or antimiR23b HBMEC cultures and measured effects on TEER. If Wnt/catenin signaling is required for the protective BBB phenotype, the inhibitor should attenuate the antimiR23b-mediated rescue of barrier properties (Figure H, right panel in red). The rescue experiments show that the miR/antimiR23b effect is partly dependent on Wnt/catenin signaling. AntimiR23b-enhanced TEER was significantly but partially reduced by the Wnt inhibitor (XAV939), and miR23b-induced loss of TEER was partially rescued by Wnt3a. Together, these orthogonal gain and loss-of-function manipulations establish Wnt/catenin signaling as a partial but necessary mediator of the protective barrier function of antimiR23b. Collectively, our findings demonstrate that miR23b and antimiR23b regulate barrier properties by modulating the Wnt/catenin signaling pathway, specifically by targeting catenin, LEF1, and key downstream effectors in BECs. Furthermore, by orchestrating this pathway alongside the TGF, Notch, VEGF, and IL6 networks, antimiR23b reinforces barrier integrity, whereas miR23b diminishes it. Future studies will determine the specific contributions of these pathways to miR23b's role in BEC and BBB biology. However, our findings establish that miR23b and antimiR23b are important regulators of Wnt/catenin signaling in BECs. By enhancing catenin, LEF1, and downstream effector proteins, antimiR23b drives a protective, barrier-enhancing phenotype. Furthermore, antimiR23b affects additional key signaling cascades, including TGF, Notch, VEGF, and IL6, which are inversely regulated by miR23b.

## 2.6 Unbiased Analysis of miR/antimiR23b Effects on the Transcriptome and Proteome of HBMEC

To provide a global unbiased view of genes, proteins, and pathway affected by miR23b/antimiR23b in HBMECs, we next generated transcriptomic and proteomic profiles comparing control, miR23b, and antimiR23b-overexpressing HBMECs to examine global transcriptional and translational changes elicited by miR23b and antimiR23b. In brief, miR/antimiR/control miR HBMECs were generated, expanded, and the barrier phenotype was validated by TEER analysis. Cells were harvested, RNA was subjected to RNA sequencing, and protein was subjected to proteomic analysis from parallel cultures. First, principal component analysis (PCA) of control vs miR23b and antimiR23b HBMEC gene profiles was performed, demonstrating separation of the three conditions (Figure 6A). One antimiR23b sample was identified as an outlier. The outlier sample was included in all differential gene expression (DGE) analyses but was excluded from the heatmaps shown in Figure 6D-G. Heatmaps including all samples are shown in Figure S5.2. FIGURE 6 antimiR23b and miR23b HBMEC transcriptome and proteome analysis demonstrate changes in genes and proteins regulating angiogenesis and barrier properties. (A) Principal component analysis (PCA) plot illustrating control vs miR23b versus antimiR23b HBMEC gene profiles. One antimiR23b sample was determined. The outlier sample was included in all DEG analysis but was removed from the shown heatmaps. (B) Volcano plot highlighting upregulated and downregulated BBB-specific genes in antimiR23b compared to miR23b HBMECs. All listed genes are significant (Padj 0.05; |logFC| 0.25). (C) Gene Set Enrichment

(GSEA) analysis pathways in anti-miR23b compared to miR23b HBMEC. Bars indicate  $\log(\text{false discovery rate})$  values. (DG) Heatmap of BBB-specific genes (TJ [tight junction] proteins, receptor-mediated transcytosis, canonical Wnt/catenin signaling and Angiogenesis between anti-miR23b and miR23b HBMECs (one sample was an outlier, see PCA plot in A, and was removed, but Heatmaps with all samples are shown in Figure S6.2). The scale bar is  $\log(Z \text{ score})$ , where dark colors indicate a high Z score and light colors indicate a low Z score. (H) PCA plot illustrating control vs miR23b versus anti-miR23b HBMEC protein profiles. (I) Volcano plot highlighting upregulated and downregulated BBB-specific proteins in anti-miR23b compared to miR23b HBMECs. All listed proteins are significant ( $P_{\text{adj}} 0.05$ ;  $|\log FC| 0.6$ ). Next, differential gene expression analysis was carried out, and the results were visualized in a volcano plot (Figure 6B), showing the upregulated and downregulated genes, including some of the BEC-specific genes in anti-miR23b compared to miR23b (Figure 6B). These analyses show that miR23b upregulates genes related to angiogenesis (e.g., CXCR4, ACKR3, ANGPT2, FZD8, CD34, SPARC, KDR/VEGFR2, PGF, VEGFA), promoting endothelial cell migration, angiogenesis, vascular destabilization, and permeability, whereas anti-miR23b upregulates some BBB-specific genes (e.g., TGFB1, NOTCH4, BRCA1, APLNR, FZD7, PGARP, KRT7), promoting vascular stability and BBB maintenance. These findings align with the concept that downregulation of angiogenic genes strengthens the BBB by shifting endothelial cells from a proliferative, high-permeability state to a quiescent, stable state, thereby allowing tight junction maturation. Reducing angiogenic signaling, particularly VEGF and components of the Wnt pathways, prevents the formation of leaky vessels and enhances barrier integrity. The results from the DGE analysis were then used for Gene Set Enrichment Analysis (GSEA) to assess pathways regulated in anti-miR23b samples compared with miR23b samples. anti-miR23b HBMECs showed upregulated genes relevant to noncanonical Wnt signaling and antigen processing and presentation, whereas miR23b samples showed upregulated genes relevant to Notch and TGF signaling, cell migration, angiogenesis, and CNS/EC cell identity (Figure 6C). These results strengthen the finding that miR23b enhances angiogenesis and CNS/EC cell identity, whereas anti-miR23b enhances Wnt signaling and antigen processing and presentation, which is critical for maintaining barrier stability and immune surveillance at the BBB. GSEA analysis of control versus miR23b DEGs confirmed that miR23b promotes angiogenesis, whereas analysis of anti-miR23b versus control confirmed suppression of angiogenic pathways and upregulation of noncanonical Wnt signaling (Figure S5.1A,B). Additionally, we generated heatmaps of BBB-specific genes (TJ proteins, receptor-mediated transcytosis, canonical Wnt/catenin signaling, and angiogenesis) in anti-miR23b and miR23b HBMECs (Figure 6DG), to demonstrate specific genes that are either up or downregulated by miR23b and anti-miR23b. Anti-miR23b treatment upregulated a subset of BBB transcriptome related to tight junctions, transcytosis, transporters and extracellular matrix proteins (Figure 6D,E, Figure S5.1C,D and Extended Data Set S2). In addition, a subset of the Wnt/catenin signaling genes are upregulated in anti-miR23b samples, whereas numerous angiogenesis genes are downregulated by anti-miR23b compared to miR23b (Figures S5.1CF, S5.2AH and Extended Data Set S2). Finally, anti-miR23b upregulated a subset of genes related to CNS/EC identity, and noncanonical Wnt signaling compared to miR23b (Figure S5.1E,F). All heatmaps, including the outlier sample, are included in Figure S5.2AH, along with gene lists of miR/anti-miR23b and control vs miR or anti-miR23b in Extended Data Set S2). Because miRs affect both mRNA expression and protein translation, proteomic analysis was performed on parallel HBMEC samples. Proteomic PCA plots between control vs miR23b vs anti-miR23b HBMEC protein profiles demonstrated clear separation among conditions (Figure 6H). Volcano plots visualizing the up and downregulated proteins in anti-miR23b compared to miR23b (Figure 6I and Extended Data Set S3), confirmed upregulation of several proteins (e.g., SERPINEB2, ADAMTS4, TGFB1, ITGB3) known to regulate barrier integrity and promote angiogenesis and inflammation. In contrast, very few proteins were upregulated by anti-miR23b overexpression (e.g., MGP, CD34, ALDH1A1, PLSCR4, FGF2) (Figure 6I and Extended Data Set S3). Some of these proteins support BBB integrity, stabilize vascular structures, and are neuroprotective and anti-inflammatory. Control vs miR23b or control vs anti-miR23b volcano plots also confirmed similar findings (Figure S5.3A,B). The proteomic analysis validated some molecules identified as miR/anti-miR23b regulated genes in the transcriptomic analysis (ITGB3, CNTNAP1, NAPG, SERPINB2, and ADAMTS4). This confirms that changes in these specific transcripts are reflected in corresponding changes at the protein level. In summary, transcriptomic and proteomic analyses reveal opposing roles for miR23b and anti-miR23b in BECs: miR23b potentially induces angiogenesis-specific genes and proteins, whereas anti-miR23b reverses this effect, upregulating a subset of BBB-specific markers. Moreover, several proangiogenic pathways (e.g. Notch, TGF) were suppressed in anti-miR23b samples to promote a functional BBB. Integrating eCLIPseq results with transcriptomic and proteomic signatures demonstrates that miR23b/anti-miR23b regulates a subset of the large network of BBB genes and functions, with critical implications for endothelial health and disease.

## 2.7 Anti-miR23b Reduces BBB Deficits in a 3D In Vitro Stroke-Like Model

Ischemic stroke induces BBB dysfunction through a cascade of cellular events, including serum protein leakage, immune cell infiltration, and inflammation, that exacerbate neuronal damage [ 68 , 69 , 70 ]. Even subtle increases in BBB permeability can lead to hippocampal injury and cognitive decline [ 71 , 72 , 73 , 74 , 75 ].

However, the underlying mechanisms of chronic BBB dysfunction after ischemic injury remain incompletely understood. We used our established 3D HBMEC culture platform to model ischemic stroke in vitro using the oxygenglucose deprivation (OGD) paradigm. Baseline cultures of both anti-miR23b and control HBMECs were established as leaktight 3D tubules before OGD exposure (days 710). 3D HBMEC tubules were then exposed to hypoxia (1% O<sub>2</sub>), glucose deprivation, and static flow for 6 h, then returned to normoxic, glucose-replete conditions with restored flow (Figure 7A). This setup enabled us to assess both acute injury, and longer-term recovery mechanisms related to BBB integrity. TEER analysis showed that while both groups exhibited a decline in barrier function during OGD, the loss was significantly less severe in anti-miR23b cultures. Furthermore, anti-miR23b cultures showed enhanced recovery following stroke-like injury (Figure 7A), suggesting a protective role for anti-miR23b in injury resistance and barrier repair. BI assays further confirmed reduced BBB leakiness in anti-miR23b cultures at both early (5 and 8 h) and later timepoints (3 and 14 days) post-stroke-like conditions (Figure 7B). **FIGURE 7 anti-miR23b reduces BBB deficits in a 3D in vitro stroke model.** (A) anti-miR23b or control HBMECs were seeded in the 3D Mimetas device. When 3D tubules were intact, cultures were exposed to in vitro stroke-like conditions (1% hypoxia, no glucose media, no/low flow). Continuous 3D TEER analysis shows partial loss of barrier function over time (n = 10 chip cultures/condition). After 6 h of in vitro stroke-like conditions, cultures were rescued (moved to normoxia) with glucose, and perfusion was reinstated, allowing for examination of resilience to in vitro stroke-like conditions and BBB repair. (B) Measurements of tubule permeability were obtained using BI assays before, during, and following in vitro strokes using the C10 confocal imager. Representative images are shown for baseline, 5, 8, 72 h, and 2 weeks after in vitro stroke-like conditions n = 5, (right panels), and quantifications of relative permeability (left graph) (n = 10 chip cultures/condition/timepoint). Scale = 250  $\mu$ m (C) Tight junction morphology was analyzed by ZO1 staining using the C10 confocal imager. The top panels show examples of intact tight junctions, junctions with zigzags, and junctions with gaps. Representative TJ morphology of anti-miR23b or controls (left panel) on day 21 following in vitro stroke-like conditions (left) and quantifications (right), n = 3 chips per condition. Scale = 20  $\mu$ m. (D) Measurement of transcytosis of anti-miR23b versus control HBMECs at 5 h post in vitro stroke-like conditions. Shown are images of albumin594 (red) transcytosis in chip images (left) and quantifications (middle left), and individual images of PLVAP, albumin594 (middle right), and quantification (far right), n = 5. Scale = 12.5  $\mu$ m. (E) Activation of Wnt/catenin post in vitro stroke-like conditions was examined at baseline, at subacute (3 days), and at chronic time point (21 days) of catenin and LEF1/DAPI, magnified images and quantifications (left panels) and tubule images (right panel), using the C10 confocal imaging plate reader, n = 5. Scale = 25  $\mu$ m for cells and 50  $\mu$ m for tubules. Graphs are shown as Mean  $\pm$  SEM of n = 310 independent biological replicates, \* p 0.05, \*\* p 0.01, and \*\*\* p 0.001, by twoway ANOVA for multiple comparisons, twotailed Student's t test when analyzing two groups, and multiple unpaired t tests for each TEER timepoint. Next, we examined TJ morphology in HBMEC 3D tubules under stroke-like conditions. Although total expression and localization of Claudin5, Occludin, and ZO1 were not significantly altered during OGD, structural analysis revealed fewer abnormal ZO1 + TJs with zigzags and gaps (green) in anti-miR23b cultures compared with controls 21 days poststroke (Figure 7C). These data suggest that anti-miR23b preserves TJ morphological architecture even under hypoxic challenge. Because stroke is known to increase transcytosis [ 61 ], we measured albumin uptake using albuminAlexa594. anti-miR23b HBMECs showed significantly reduced albumin uptake 5 h postOGD relative to controls (Figure 7D). IF analysis also demonstrated decreased levels of PLVAP and intracellular albumin in anti-miR23b cultures (Figure 7D), confirming suppression of transcytosis under ischemic conditions. To determine whether these are downstream of Wnt/catenin activity, we measured catenin and LEF1 protein levels in anti-miR23b and control cultures before and after OGD. LEF1 expression was modestly reduced in both groups at 3 days poststroke. By day 21, anti-miR23b cultures showed significantly higher expression of both catenin and LEF1 than controls (Figure 7E). These findings suggest that anti-miR23b not only preserves baseline Wnt/catenin signaling but also promotes its reactivation during later stages of BBB recovery. In summary, anti-miR23b preserves BBB integrity in a 3D stroke model by accelerating barrier recovery and reducing breakdown. It does so by maintaining TJ protein structure, suppressing harmful transcytosis, and bolstering Wnt/catenin signaling to restore endothelial barrier function after stroke.

## 2.8 Anti-miR23b Enhances Tight Junction Structure and Reduces Stroke Volume and Outcome Measures in Proof-of-Concept Experiments Using an In Vivo Ischemic Stroke Model

Ischemic stroke causes irreversible damage to the infarct core and compromises the surrounding penumbra, a region considered therapeutically salvageable. BBB disruption is an early pathological feature that contributes to the conversion of the penumbra into the infarct core and is associated with poor clinical outcomes [ 76 , 77 , 78 ]. Because stroke-induced BBB disruption contributes directly to infarct expansion and cognitive decline, we asked whether anti-miR23b could confer protection in a clinically relevant in vivo model of ischemic stroke. To assess the in vivo relevance of our in vitro findings, we conducted proof-of-concept studies using the transient middle cerebral artery occlusion (tMCAO) model to test whether prophylactic treatment of the brain endothelium with anti-miR23b could reduce BBB dysfunction and stroke severity. This stroke model induces transient cerebral ischemia in



the cortex and striatum by inserting an intraluminal suture into the internal carotid artery [ 61 ]. To selectively target brain ECs, we used an adenoassociated viral vector (AAVBR1) developed by the Körbelin lab [ 79 ]. To accurately assess the effects of anti-miR23b intervention, we employed a miRcontrolAAVBR1 vector as an experimental control, as previously described [ 80 , 81 , 82 , 83 , 84 , 85 ]. Given that the tMCAO surgical procedure and vector capsid are consistent across both groups, any phenotypic or molecular differences can then be ascribed to the anti-miR23b cargo. A single intravenous injection (10 1 1 gp/mouse) delivered retroorbitally resulted in robust transduction efficiency ( 75%) in the brain endothelium, as confirmed by costaining with GFP (from AAVBR1) and Podocalyxin, a vascular marker (Figure 8A ). In contrast, astrocytes (GSpositive) and neurons (NeuNpositive) were targeted at low levels ( 2%) by the AAVBR1 viral vector (GFP) in our system (Figure 8A and Figure S6A ), validating brain endothelial cell specificity, as previously shown [ 79 , 80 , 85 , 86 , 87 ]. Retroorbital injection rapidly delivers therapeutic vectors to the ophthalmic artery and cavernous sinus, saturating the cerebrovascular bed for widespread BBB transduction. This route limits systemic exposure because the BBB endothelium clears the vector, which spares sensitive neurons from parenchymal toxicity. The transgene cassette used affects the number of nonECs expressing the AAVBR1delivered cargo, which may explain why some studies report neuronal infection [ 88 ]. The treatment was well tolerated, with mice maintaining stable body weight and showing no significant systemic immune activation, as confirmed by serum cytokine profiling (IL1, IL6, IFN, TNF, IL8, IL10, or IL12) up to 3 months postinjection (data not shown). Consequently, AAVBR1 represents a promising platform for brain endothelialspecific delivery of anti-miR23b.

**FIGURE 8 Brain endothelial anti-miR23b overexpression before stroke reduces acute BBB permeability, enhances tight junction formation, and reduces poor outcome measures, following tMCAO.** (A) Mice were transduced with AAVBR1 anti-miR23b or control miR viruses before stroke to target endothelial cells. A representative image of AAVBR1controlGFP (green) virus showing expression in brain capillaries costained with podocalyxin (PODXL, red vessel marker) left panel, glutamine synthetase (GS, red, astrocytic marker) middle panel, or Neuronal Nuclei (NeuN, red, neuronal marker) right panel, three weeks postinfection, n = 3. Scale = 100 and 50  $\mu$ m (GS and NeuN). (B) miR23b specific qPCR analysis was performed on total brain RNA from PFA sections of control tMCAO mouse brain, comparing normal versus stroke regions. miR23b expression levels were normalized to RNU1A, n=3 mice per condition. Twoway ANOVA, bars represent. Bar shows mean  $\pm$  SEM. (C,D) 20weekold male mice were injected with anti-miR23bAAVBR1 or control miRAAVBR1 7 days before the tMCAO surgery (75minute occlusion). Some mice were sacrificed 4 days after tMCAO), and BBB damage was examined by analysis of serum IgG extravasation (green). The leakage volume (left panel), average leakage area (middle panel), and average leakage intensity (right panel) for serum IgG extravasation ( n = 5 mice/group) are shown, Scale = 1 mm. Unpaired student t tests for volume and at each Bregma section for average leakage section and intensity across mice. Paired student t test for each mouse when comparing contralateral versus ipsilateral sections for average intensity. Bars and XY graphs show Mean  $\pm$  SEM. (E) Some mice were exposed to tailvein injections of a small tracer [biocytinTMR (red); 890 Da] for 45 min before tissue harvest to visualize BBB permeability. BiocytinTMR leakage volume (left panel), average leakage area (middle panel), and average leakage intensity (right panel, n = 3 mice/group), Scale = 1 mm. Unpaired student t tests for volume and at each Bregma section for average leakage section and intensity across mice. Paired student t test for each mouse when comparing contralateral vs ipsilateral sections for average intensity. Bars and XY graphs show Mean  $\pm$  SEM. (F) Neurological scoring was monitored daily. A poor neurological score of 5 is shown. (G) Survival monitoring results are depicted in ( n = 11 mice/group). (H) Analysis of TJ protein Claudin5 expression of Glut1 positive brain capillaries was performed comparing control versus anti-miR23b AAVBR1 tMCAO tissue, 3 days post injury. Representative images are shown (left panels), and quantification of tight junction strand structure is shown in the right panel. Intact TJ strands (white arrows), Absent TJ strands (red arrows), and TJ strands with Gaps (yellow arrows) in vascular area of the ipsilateral region of stroke were quantified, n = 3 mice per condition. Scale = 100  $\mu$ m. Twoway ANOVA. Bar shows mean  $\pm$  SEM. (I) LEF1 expression was examined in CD31 positive vessels of control vs anti-miR23b AAVBR1 treated mice. White arrow represents LEF1+ brain endothelial cells in CD31+ vessels, n = 4 mice per genotype. Scale = 100  $\mu$ m. Student t test. Bar shows Mean  $\pm$  SEM. To evaluate therapeutic potential, 34monthold male mice were randomized and injected with AAVBR1anti-miR23b or AAVBR1controlmiR one week before 75 min tMCAO. Physiological parameters, including cerebral blood flow, temperature, and heart rate, were monitored in a blind manner, and immune status was assessed using plasma samples (data not shown). To assess miR23b expression in the brain, we performed in situ hybridization on healthy and tMCAO mouse brains. We found that miR23b mRNA was highly upregulated in many cells in the stroke border region, which is a region where angiogenic remodeling of the vasculature is increased together with endothelial Wnt/catenin activity (Figure S6B ). To validate these findings, miRspecific qPCR was performed on total brain tissue from control and tMCAO subjects. This analysis confirmed significantly elevated levels of miR23b and miR126, with modestly increased levels of miR21 in strokeaffected brains relative to control tissue (Figure 8B ). In summary, the in situ and qPCR data suggest poststroke upregulation of

miR23b, consistent with reported increases in plasma miR23b levels in stroke patients [ 89 ]. Consequently, targeting miR23b with an anti-miR23b competing RNA may provide therapeutic benefits after a stroke. A subset of mice was sacrificed 96 h post-tMCAO to assess acute outcomes. The extent of BBB leakage was evaluated by both serum IgG extravasation and biocytinTMR (870 Da) tracer leakage administered intravenously 45 min before euthanasia (Figure 8C ). Control mice showed extensive BBB leakage in the cortex. In contrast, prophylactic treatment with AAVBR1-anti-miR23b significantly reduced both the area and intensity of serum IgG and biocytinTMR leakage (Figure 8D,E ). Quantitative analysis revealed twofold reductions in total BBB leakage area across several bregma regions (Figure 8D,E ) and approximately twofold reductions in serum IgG or tracer intensity in anti-miR23b-treated mice compared to controls (Figure 8C,E ). Importantly, post-tMCAO barrier integrity analysis (via tracer/IgG extravasation) recapitulated the tracer/stroke volume results from our in vitro studies. Alongside tMCAO stroke volume analysis, we evaluated neurological scores and survival rates in mice. Compared with controls, the anti-miR23b group had a lower percentage of animals with severe neurological impairment (score 5) and higher survival rates (Figure 8F ). These findings suggest that anti-miR23b provides at least some acute neuroprotection (Figure 8G ). To further evaluate the in vivo relevance of our in vitro stroke model, we assessed Claudin5 expression in GLUT1-positive brain capillaries in tissue from control versus AAVBR1-anti-miR23b-treated mice after tMCAO. Anti-miR23b treatment resulted in a significant increase in intact TJ strands and a corresponding reduction in absent TJ protein strands compared to controls (Figure 8H ); nonetheless, localized gaps in TJ strands persisted in a subset of the anti-miR23b group. Post-tMCAO, we also assessed LEF1 expression in CD31-positive brain vessels, comparing control and anti-miR23b-treated mice. No significant difference was observed between the groups (Figure 8I ). This aligns with our in vitro stroke model results, where LEF1 and catenin protein levels showed only modest, nonsignificant increases at 3 days post in vitro stroke, in contrast to day 21 post stroke, when both proteins were significantly upregulated. Despite a limited sample size, we could recapitulate key in vitro findings in our in vivo stroke model, demonstrating that anti-miR23b modulates BBB permeability (stroke volume) and TJ protein structure. However, our current findings are limited by a small sample size, the use of only two functional tracers, and a primary focus on short-term stroke outcomes. Future studies will employ a broader range of tracers (e.g., Evans blue, Dextran) to evaluate BBB preservation. Furthermore, these studies will incorporate post-stroke angiogenesis and longitudinal behavioral assessments to validate the translational efficacy of anti-miR23b in restoring BBB function following ischemic stroke. In summary, our preliminary in vivo findings indicate that prophylactic treatment with AAVBR1-delivered anti-miR23b effectively may mitigate BBB leakage and reinforce tight junction structure. Further studies are needed to assess the effects of anti-miR23b in broader stroke outcomes in the tMCAO stroke model to determine whether anti-miR23b may be a promising therapeutic strategy for ischemic stroke.

### 3 Discussion

Although acute BBB dysfunction is recognized as one of the earliest and most detrimental events in ischemic stroke [ 5 , 6 , 7 ], a definitive understanding of its mechanisms, timing, and optimal treatment strategies for both protection and repair remains elusive. Our findings establish that targeted silencing of miR23b in the brain endothelium enhances BBB resilience to injury and facilitates repair through both Wnt/catenin-independent and independent mechanisms. Inhibiting miR23b in BECs strengthens the BBB by upregulating critical junction proteins (ZO1, Claudin5, Occludin, and VECadherin) and reducing PLVAP-mediated transcellular transport. These protective effects, observed in both healthy and ischemic environments, significantly improve the structural and functional integrity of the BBB. Conversely, overexpression of miR23b impairs barrier function, confirming its role as a negative regulator of BBB stability. We developed an unbiased anti-miR screening strategy and identified anti-miR23b/miR23b as a conserved anti-miR/miR with potent regulatory activity of endothelial barrier properties, including barrier integrity. Other miRs, including miR129, miR221, and miR191, have also been implicated in regulating vascular responses in stroke pathology, including angiogenesis, highlighting a complex miR regulatory network governing BBB function in neurological diseases [ 72 , 73 , 74 ]. However, these miRs were not identified in our anti-miR screening strategy targeting regulators of endothelial barrier integrity. miR-specific qRT-PCR analyses revealed that miR23a, miR23b, and miR27 (which belong to the same miR clusters [ 54 ]) are highly expressed in young mouse BECs. However, their levels are reduced in aged BECs, even though they remain highly expressed, suggesting that downregulation of their expression may also be a critical event in promoting the acquisition and maturation of barrier properties of BEC. Our comprehensive mechanistic study confirms the hypothesis that miR23b is an endogenously expressed miR that negatively regulates barrier integrity in both mouse and human BECs. miRs can function either by directly binding to and suppressing the translation of specific mRNAs or by targeting mRNAs that influence broader signaling pathways [ 90 , 91 ]. Our multipronged approach to assess whether the observed miR/anti-miR effects stem from direct versus indirect regulation demonstrates that miR23b directly targets VECadherin/CDH5, catenin/CTNNB1, and LEF1 mRNA through imperfect seed interactions, whereas Claudin5, Occludin, and PLVAP appear to be regulated indirectly through endothelial Wnt/catenin activity. Surprisingly, ZO1/TJP1 was not identified as a direct target in miR eCLIPseq analysis. However, reporter assays and bioinformatic analyses

suggest that miR23b may bind TJP1 mRNA under certain cellular conditions. We adopted the unbiased miR23b eCLIPseq approach to further examine immediate/early functional outputs elicited by miR23b or anti-miR23b in BECs, showing that anti-miR23b functions as a master regulator of BEC by controlling key signaling pathways (TGF, VEGF, Notch, IL6), in addition to the Wnt/catenin pathway. The Wnt/catenin, TGF, and Notch signaling pathways cooperate to maintain barrier stability and prevent leakiness. In contrast, VEGF and IL6 signaling generally disrupt the BBB but can also promote beneficial angiogenesis and neuroprotection following injury. To determine whether miR23b directly affects Wnt/catenin activity, we performed a WntGFP reporter assay. We then used Wnt/catenin rescue experiments to evaluate the Wnt-dependent and Wnt-independent contributions to anti-miR23b barrier function, demonstrating that barrier function is partially Wnt-dependent. We leveraged multiomics datasets to critically assess the multipronged role of anti-miR23b in protective barrier function: first, early/intermediate outputs identified by miReCLIPseq analysis showed direct effects on BBB/TJ formation, cell-cell adhesion, transcytosis, and ECM, as well as global effects on Wnt/catenin signaling pathways, modulating numerous key mRNAs/proteins (including CTNNB1, LEF1, WNT7A, FZD4, LRP5, LRP6, APC, TCF3, TCF4, DKF1, WNT5A, ROCK1, DVL2, DVL3, and DAAM1). Additionally, anti-miR23b regulates the TGF and Notch pathways, which integrate with Wnt signaling to maintain barrier stability. In contrast, VEGF and IL6 pathways, also regulated by anti-miR23b, typically increase BBB permeability but can also promote protective angiogenesis and neurogenesis. The integrative multiomics analysis robustly validated that miR23b is a potent inducer of angiogenesis-related pathways in BECs, while anti-miR23b effectively inhibits angiogenesis genes and pathways and strengthens a subset of BBB gene signatures. These findings align with previous studies showing that reducing proangiogenic pathways including Wnt/catenin signaling in BECs is essential to promote and maintain BBB integrity. Our detailed profiling of miR23b and its inhibitor in BECs underscores the context and tissue-specific activities of miR23b, shedding new light on the seemingly contradictory roles of miR23 in angiogenic regulation [ 54 , 89 , 90 ]. Our results indicate that inhibiting miR23b may be beneficial in stroke models, which contrasts with evidence that its induction promotes neuroprotection in Alzheimer's disease and traumatic brain injury [ 42 , 45 , 95 , 96 , 99 , 100 , 101 , 102 ]. These disparities likely stem from distinct, cell-type-specific targets of miR23b, underscoring the need for nuanced, context-dependent approaches to miR-based therapeutics. Although our analysis focused on anti-miR23b-mediated inhibition in BECs, potential cross-reactivity with miR23a, due to their single-nucleotide difference, cannot be ruled out. However, we anticipate minimal impact from inhibiting both variants, given their substantial target overlap. Future studies will confirm this assumption. Several miRs have already been implicated in regulating stroke outcomes. miR15a and miR124 have been associated with ischemic stroke progression and protection, respectively [ 77 , 80 , 85 , 88 ]. miR129, miR221, and miR191 regulate vascular responses during stroke pathology [ 89 , 92 , 93 ]. Our data expand on previous reports showing that global, nonspecific lentiviral inhibition of miR23b alleviates ischemic injury by enhancing neuronal survival through the Nrf2 and cMyc signaling pathways [ 102 , 103 ]. Interestingly, elevated serum levels of miR23b3p are observed in patients with ischemic stroke and transient ischemic attack, suggesting a potential role in hypoxia-induced BBB disruption [ 89 ]. Yet miR23b also promotes myelination and neuroprotection [ 104 ], highlighting a complex and paradoxical role for miR23b across different CNS cell types. Our findings from both the in vitro model under stroke-like conditions and the in vivo stroke model (tMCAO) using a brain endothelial-specific approach provide significant insight into the mechanisms by which miR23b inhibition may confer enhanced BBB stability and possible neuroprotection in ischemic stroke. With miR23b serving as a negative regulator of BBB integrity, inhibition of miR23b protects the BBB by enhancing resistance to injury and facilitating repair. It acts by modulating BBB-specific proteins and activating key pathways, such as the Wnt/catenin pathway, to promote BBB structural stability and recovery. One potential limitation of anti-miR therapeutics concerns target specificity. Should the payload fail to distinguish perfectly between targets, it may inadvertently silence unintended homologous mRNAs, leading to unexpected phenotypic alterations. Therefore, additional studies are required to comprehensively assess off-target effects induced by anti-miR23bAAVBR1. Additionally, current limitations of our study include a small sample size, reliance on two functional tracers, and the exclusive assessment of short-term functional outcomes. Future research will utilize a wider array of tracers (e.g., Evans blue, Dextran) to track BBB integrity. Additionally, evaluating poststroke angiogenesis and longitudinal behavioral responses will help establish the translational relevance of anti-miR23b-mediated BBB restoration after ischemic stroke. A major bottleneck to understanding BBB pathology in stroke is the lack of human-relevant, high-fidelity in vitro models. Current in vivo models, including the tMCAO model, are valuable but limited by species-specific differences and labor-intensive methods that prevent extensive drug screening [ 97 , 98 ]. Our 3D microfluidic system, bridging simpler 2D assays and complex in vivo models, serves as an ideal intermediate platform for studying stroke pathology and screening therapeutic candidates. Our 3D microfluidic BBB model significantly improves on traditional 2D static cultures, providing a robust system for studying mechanisms of ischemic stroke. Our studies demonstrate the value of this 3D BBB culture model as a predictive and mechanistically robust platform,

enabling detailed exploration of brain endothelial responses to ischemic injury and facilitating highthroughput screening of nextgeneration therapeutics. Consistent with our 3D BBB model findings, our in vivo study shows that targeting BECs with anti-miR23b preserves tight junction structure, reduces vascular leakage and stroke volume, and ultimately improves outcomes in ischemic stroke. While anti-miR23b shows promise as a therapeutic avenue for ischemic stroke, future enhancements should focus on increasing physiological relevance. Integrating a more comprehensive NVU/CNS model that incorporates BBB-supporting cells (pericytes and astrocytes) alongside neurons and microglia will be critical to advancing this therapeutic potential. The therapeutic potential of anti-miR strategies is illustrated by existing miR-targeted therapies, such as anti-miR122 (Miravirsen), which has proven safe and effective against Hepatitis C [ 105 , 106 , 107 , 108 , 109 ]. Advanced delivery strategies, such as circular anti-miRs or nanoparticle-based approaches, could enhance therapeutic specificity and efficiency for CNS targeting. There is also significant interest in identifying Wnt/catenin agonists that restore homeostatic signaling without overstimulation to promote BBB repair, particularly after stroke. Recently, Wnt7a has been engineered to interact with Grp124/Reck, selectively targeting BEC to promote Wnt/catenin signaling and BBB repair in ischemic stroke [ 17 ]. Targeting endogenous BEC molecules, such as miR23b, may offer another safe and effective strategy to achieve this therapeutic goal. Our proof-of-concept studies employing targeted anti-miR23b delivery via AAVBR1 specifically improved BBB integrity and outcomes in the tMCAO model, suggesting clinical potential. Future research will explore poststroke therapeutic interventions using anti-miR23b, including evaluations in middle-aged and aged mice of both sexes. These studies will comprehensively assess neuronal injury, inflammation, immune cell infiltration, and long-term behavioral and cognitive outcomes, thereby significantly advancing translational potential. Ultimately, RNA-based neuroprotection, such as anti-miR23b, offers superior long-term therapeutic potential compared with conventional pharmacological interventions, which are severely limited by short systemic half-lives. Localized delivery techniques, specifically catheter-based administration during mechanical thrombectomy, could optimize clinical application and significantly improve repair of neurovascular integrity. In conclusion, given that BBB dysfunction is a critical factor in aging and in neurological conditions, including TBI, MS, Alzheimer's, and NeuroHIV, our findings represent a significant advance toward therapeutic, small RNA-based gene therapy strategies to restore BBB integrity across multiple CNS pathologies.

#### 4 Materials and Methods

##### 4.1 Cell Culture

Mouse lung endothelial cells (mLEC; C576011, Cell Biologics, Chicago, IL, USA) were maintained in complete mouse EC media (M1168, Cell Biologics, Chicago, IL, USA) and 10% FBS (FB02, Omega Scientific, Tarzana, CA, USA). Human lung EC (human LEC; #3000, ScienCell, Carlsbad, CA, USA) were maintained on plates coated with 10  $\mu$ g/mL fibronectin (F2006, SigmaAldrich, St. Louis, MO, USA) in EC media (1001, ScienCell). Primary human Brain Microvascular Endothelial Cells (HBMEC, ACBRI76, Cell Systems, Kirkland, WA, USA) were cultured on 15  $\mu$ g/mL fibronectin coated T75 flasks (353112, Corning, Corning, NY, USA) in EndoGroMV Complete Medium (SCME004, SigmaAldrich, St. Louis, MO, USA) supplemented with 5 ng/mL bFGF (3718FB025, R D Systems, Minneapolis, MN, USA). Human Cerebral Microvascular Endothelial Cell line (hCMEC/D3; SCC066, SigmaAldrich, St. Louis, MO, USA) cells were grown on Rat Collagen Type I (C38671VL, SigmaAldrich, St. Louis, MO, USA) coated T75 flasks in EndoGroMV complete Medium supplemented with 5 ng/mL bFGF. Primary Human Umbilical Cord Vein Endothelial Cells (HUVEC, PCS100010, ATCC, Manassas, VA, USA) were cultured on 1% Gelatin (G9136, SigmaAldrich, St. Louis, MO, USA) coated T75 flasks in Endothelial Cell Growth Medium MV2 (C22022, PromoCell, Heidelberg, Germany). HBMECs and HUVECs were used between passages 4 and 8. C57BL/6 Mouse Embryonic Brain Endothelial Cells (C576023E, Cell Biologics, Chicago, IL, USA) and Aged Mouse Brain Microvascular Endothelial Cells (A576023, Cell Biologics, Chicago, IL, USA) were cultured on 1% gelatin-coated T75 flasks and grown in complete Endothelial Cell Medium (1001, ScienCell, Carlsbad, CA, USA). HEK293T cells (CRL3216, ATCC) were cultured on 1% gelatin-coated flasks in DMEM/F12 medium supplemented with 10% FBS. All cells were cultured at 37°C and 5% CO<sub>2</sub>. The Wnt inhibitor XAV939 (5  $\mu$ M, 3748, R D Systems, Minneapolis, MN, USA) or Wnt agonist Wnt3a (100 ng, 5036WN, R D Systems, Minneapolis, MN, USA) were added to some cultures.

##### 4.2 Cell Transductions and Transfections

**Stable transduction of miRs:** VSVG-pseudotyped lentiviral particles were made by transfecting 5.3  $\mu$ g of pMD2G (12259, Addgene, Watertown, MA, USA), 9.7  $\mu$ g of pCMVDR8.74 (8455, Addgene, Watertown, MA, USA), and 15  $\mu$ g of pCD510B1 (miR Control), mZIP, pCD510B1-miR23b, or mZIP-anti-miR23b into 293T cells using Lipofectamine LTX (15338030, ThermoFisher, Waltham, MA, USA). Viral supernatants were concentrated using PEGit (LV810A1, System Biosciences, Palo Alto, CA, USA). Cells were transduced with high-titer virus using polybrene (SC134220, Santa Cruz Biotechnology, Dallas, TX, USA) and spinoculated at 800  $\times$  g at 32°C for 30 min. HBMEC and hCMEC/D3 transduced cells were selected at 48 h and maintained with 1 or 3  $\mu$ g/mL puromycin (J67236.8EQ, ThermoFisher, Waltham, MA, USA), respectively. **miR mimic transfection:** HUVECs were seeded at 300K/well in a 6-well plate and grown until confluent. Transfections were prepared using Lipofectamine RNAiMAX Transfection Reagent kit (13778030, ThermoFisher, Waltham, MA, USA) and using both Anti-miR miRNA23b Inhibitor (AM17000, ThermoFisher, Waltham, MA, USA) and Anti-miR miRNA Inhibitor Negative

Control #1 (AM17010, ThermoFisher, Waltham, MA, USA). The final miRNA concentration added to each well was 25 pmol/well, and the final Lipofectamine RNAiMAX added to each well was 7.5  $\mu$ L/well.

**4.3 AntimiR Lentiviral Library Screen** mLECs were plated at 10 K cells per well in a 96well plate, incubated with 8  $\mu$ g/mL polybrene (SC134220, Santa Cruz Biotechnology, Dallas, TX, USA) and transduced by spinoculation (800 x g at 32°C for 30 min) with the miRZip<sup>TM</sup> antimiR pooled lentiviral library (MZIPPLVA1, System Biosciences, Palo Alto, CA, USA) using 10 000 IFUs per well to ensure an MOI 1. Puromycin selection (1  $\mu$ g/mL) was initiated 48 h after transduction with full media changes every second day. copGFP expression identified cells that had been successfully infected (CopGFP (ppluGFP2) is a bright, green fluorescent protein derived from the copepod *Pontellina plumata*, known for its fast maturation and high stability). Cells were expanded, and then singlecell dilution was performed in 96well format. Single cell clones were expanded, and when confluent, cells were transferred to 96wells allowing for transendothelial electrical resistance (TEER) measurements (96W20idf PET plates, Applied BioPhysics). The plates were stabilized with 10 m m LCysteine (C6852, SigmaAldrich) coated with 100  $\mu$ g/mL PolyDlysine (P6407, SigmaAldrich), then coated with 10  $\mu$ g/mL fibronectin (F2006, SigmaAldrich). For all ECIS experiments, endothelial cells were plated at 50 000 cells/well. TEER was measured at 4000 Hz for 7296 h. For all ECIS experiments, endothelial cells were plated at 50 000 cells/well. AntimiR hits were identified by cloning and sequencing. Next, control miR, miR23b overexpression, and antimiR23b (knockdown) hightiter viruses were generated, and human lung endothelial cells (hLECs) were transduced and selected with 10  $\mu$ g/mL puromycin, after which an ECIS experiment was performed. Finally, HUVEC were transfected with control or antimiR23b oligonucleotides, and control or antimiR23b HBMEC transduced cells were challenged with TEER analysis.

**4.4 Cell Proliferation Analysis** Cells were plated at low density (6well plates at 20 K cells/well) in triplicate and counted.

**4.5 TEER Analysis** ECIS analysis: The Electrical CellSubstrate Impedance Sensing system (ECIS Z; Applied Biophysics, Troy, NY, USA) was used to monitor transendothelial electrical resistance (TEER) in real time in 96well plate format (96W20idf PET plates, Applied Biophysics, Troy, NY, USA). The plates were stabilized with 10 m m LCysteine (C6852, SigmaAldrich, St. Louis, MO, USA) coated with 100  $\mu$ g/mL PolyDlysine (P6407, SigmaAldrich, St. Louis, MO, USA), then coated with 10  $\mu$ g/mL fibronectin (F2006, SigmaAldrich, St. Louis, MO, USA). All ECIS experiments had lung endothelial cells (LECs) plated at 50 000 cells/well. TEER was measured at 4000 Hz for days to weeks.

**EVOM3 analysis:** Parental and miRmodulated HBMECs and HUVECs were both seeded on 15  $\mu$ g/mL Fibronectin or 0.1% Gelatincoated Transwell inserts in a 12well plate (3422, Corning, Glendale, AZ, USA) at 300K cells/well. TEER measurements were performed using an EVOM3 voltohmmeter (World Precision Instruments, Sarasota, FL, USA) at the start of culture and every 24 h thereafter. Resistance values (  $\Omega$  ) were multiplied by 1.13 cm<sup>2</sup>, the surface area of the Transwell insert, to obtain TEER values (  $\Omega$  x cm<sup>2</sup> ). TEER values of an empty insert were subtracted from each measurement.

**OrganoTEER 3D TEER .** HBSS was added to the middle and bottom channel inlets and outlets (50  $\mu$ L) of the OrganoPlate 3lane 40/64chip cultures, and fresh HBMECBM was added to the top channel inlets and outlets (50  $\mu$ L). The OrganoTEER device was placed on top of the OrganoPlate, and continuous TEER (  $\Omega$  x cm<sup>2</sup> ) measurements were taken every 8 min using the low TEER setting. Alternatively, snapshot TEER measurements were performed intermittently, when multiple chip culture plates were analyzed in parallel.

**4.6 RNA Isolation** Total RNA was isolated from HBMECs (ACBRI76, Cell Systems, Kirkland, WA, USA), Mouse Embryonic Brain ECs (C576023E, Cell Biologics, Chicago, IL, USA), and Aged Mouse Brain ECs (A576023, Cell Biologics, Chicago, IL, USA) following the manufacturer's protocol from mir Vana miRNA Isolation Kit (AM1560, ThermoFisher, Waltham, MA, USA). Alternatively, RNA was isolated from both contralateral and ipsilateral sides of PFAfixed mouse brain tissue by treating tissue with lysis buffer containing 0.5 mg/mL proteinase K (4333793, Invitrogen, Waltham, MA, USA) overnight at 60°C. The following day, RNA was isolated using mir Vana miRNA Isolation Kit according to the manufacturer's instructions. Total RNA was quantified using a NanoPhotometer (IMPLEN, Munich, Germany). Next, cDNA was made using miRCURY LNA RT Kit (339340, Qiagen, Venlo, Netherlands) and qPCR was carried out using the miRCURY LNA SYBR Green PCR Kit (339346, Qiagen, Venlo, Netherlands) and the following miRCURY LNA miRNA PCR Assays from Qiagen: hsamiR181d5p (YP00204789), hsamiR1495p (YP00204321), hsamiR243p (YP00204260), hasmiR95p (YP00204513), hasmiR1265p (YP00206010), hasmiR215p (YP00204230), hasmiR175p (YP02119304), hsalet7b3p (YP00205653), hsamiR23a3p (YP00204772), hsamiR27b3p (YP00205915), hsamiR23b3p (YP02119314), hsamiR29a3p (YP00204698), hsamiR2415p (YD00610842), hsamiR301a3p (YP00205601), hsamiR3753p (YP00204362), hasmiR151a3p (YP00204576) and RNU1A1 (YP00203909), on a StepOnePlus RealTime PCR System (4376600, Applied Biosystems, Waltham, MA, USA). All miRCURY LNA miRNA PCR Assays were confirmed to target human and mouse sequences. Each assay was carried out in triplicate, and gene expression for each miR was normalized to the reference gene RNU1A1.

**4.7 Western Blotting** Cells were lysed in cold RIPA buffer (89901, ThermoFisher, Waltham, MA, USA) containing protease inhibitor cocktail (PI78410, ThermoFisher, Waltham, MA, USA) to extract protein. Sample